

August 7, 2026

Dear Hiring Team,

I am writing to apply for the Data Engineer, Data Innovation position at MNP in Calgary. With a Master of Science in Computer Science from the University of Calgary and hands-on experience building Python/C++ software, SQL-backed backend services, ETL-style data pipelines, and scalable data-processing workflows, I am excited by the opportunity to contribute to MNP's Technology - Development team.

My graduate research focused on building reliable software around complex real-world datasets. Working with BigGeo, Mitacs, and the University of Calgary, I built Python and C++ pipelines for ingesting, preprocessing, storing, retrieving, and evaluating large Earth observation datasets. I automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and external data APIs, and developed DGGs-based workflows that transformed imagery into structured tensor files for downstream analysis. This work gave me practical experience with data ingestion, transformation, performance, reproducibility, and data quality.

That background aligns well with MNP's focus on modern data solutions, pipeline reliability, and hybrid data architecture. I have not worked directly with Azure Databricks, Microsoft Fabric, Unity Catalog, or Microsoft Purview in production, but I bring a strong foundation in Python, SQL, structured data workflows, APIs, Git-based development, Docker, AWS-based data coursework, and the ability to learn new platforms quickly. My research work also required careful reasoning about storage models, time-series data, multiresolution retrieval, and analytical access patterns.

I also bring backend and communication experience that would help in a collaborative data engineering team. As a back-end developer, I built Django REST APIs for NLP, text-to-speech, and speech-to-text services backed by PostgreSQL, implemented authentication flows, and worked with structured service and payment data. As a teaching assistant at the University of Calgary, I supported students in big data applications, data-at-scale workflows, SQL, AWS-based pipelines, and database design, which strengthened my ability to explain technical tradeoffs clearly to both technical and non-technical audiences.

I am particularly interested in this role because it combines practical delivery with innovation: building data pipelines, supporting Lakehouse-style architecture, working with APIs and structured or semi-structured data, and helping business teams make better use of data. I would bring ownership, careful documentation, strong learning habits, and a software engineering mindset to MNP's Data Innovation work.

Thank you for considering my application. I would welcome the opportunity to discuss how my data pipeline, Python, SQL, backend, and large-scale data-processing experience can contribute to MNP's data engineering initiatives.

Sincerely,

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